AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM356
Interaction Design and User Experience

الجامعة العربية المفتوحة
Open University
القبول والتسجيل
الامتحانات
Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 16/Apr. /2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 hours**

Instructions:

- Total Marks:**60 marks**
- This exam consists of **6 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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PART 1: CHOOSE THE CORRECT ANSWERS FOR ALL QUESTIONS [10 Marks]

1. What term describes the degree to which a product can be used by specified users to achieve specified goals with accuracy and completeness?
   a) Efficiency b) Effectiveness c) Satisfaction d) Utility
2. Which interaction principle refers to the property of an object that suggests how it should be used (e.g., a button that invites clicking)?
   a) Visibility b) Feedback c) Affordance d) Consistency
3. Which principle emphasizes the importance of providing clear and immediate feedback to user actions?
   a) Visibility b) Feedback c) Affordance d) Consistency
4. A Low-fidelity prototype, such as a series of sketches showing a user's progress through a task, is known as a:
   a) Wireframe b) Storyboard c) Wizard of Oz d) Functional Prototype
5. When a user understands how a new smartphone works because they have used a similar one before, they are utilizing:
   a) Emotional Design b) Mental Models
   c) Cognitive Bias d) System Architecture
6. In the Iterative Design process, evaluation should ideally happen:
   a) Only at the very end of the project.
   b) Only after the coding phase is complete.
   c) Throughout the entire lifecycle, starting from early requirements.
   d) Only if the client requests changes.
7. Which usability principle is violated if an app uses a Save icon in one menu but a Disk icon for the same function in another menu?
   a) Visibility b) Consistency c) Constraints d) Mapping
8. The term Utility in interaction design specifically refers to:
   a) How pleasant the app looks.
   b) Whether the product provides the right functions to the user.
   c) The speed of the internet connection is great.
   d) The price of the software license.
9. Which data gathering method involves a group of people led by a facilitator to discuss their feelings and perceptions about a product?
   a) Contextual Inquiry b) Naturalistic Observation
   c) Focus Groups d) Quantitative Surveys
10. Designing for Inclusive Design means:
    a) Designing only for people with permanent disabilities.
    b) Designing to ensure the product is usable by the widest possible range of people.
    c) Including every feature in a single interface.
    d) Designing only for the most technically advanced users.

PART 2: ANSWER ALL QUESTIONS [20 Marks]

Question 1:

A) Explain the relationship between Mapping and Feedback in interface design. Provide one unique example for each principle in a digital context. [6 marks]

B) Discuss how Visibility affects a user's ability to learn about a new system (Learnability). [4 marks]

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Question 2:

You are conducting user research to design a Campus Navigation App for new university students.

A) Describe how you would use Interviews to gather requirements. List three key questions you would ask. [4 marks]

B) Identify three types of Environmental Factors that might influence how students use this app on campus. [3 marks]

C) Discuss two potential limitations of using interviews alone as your primary data gathering method. [3 marks]

PART 3: ANSWER ALL QUESTIONS [30 marks]

Question 1: [15 marks]

"EcoHome" is a smart-home application used by families to monitor energy consumption and control home appliances. While users appreciate the energy savings, they complain that the app's interface is "overwhelming" during peak hours, alerts are often missed because they look like regular system notifications, and it takes too many clicks to perform a simple task like turning off a light.

A) Identify and explain three UX challenges that users face in this scenario. [6 marks]

B) Evaluate how these challenges impact the Efficiency and User Satisfaction of the "EcoHome" system. [4 marks]

C) Propose two specific technical or design improvements to solve these issues and justify your choice. [5 marks]

Question 2: [15 marks]

A) Explain how Cognitive Capabilities and Physical Capabilities impact the accessibility of a digital product. Provide two specific examples for each. [10 marks]

B) How does understanding these capabilities inform the creation of User Personas in the inclusive design process? Provide two ways. [5 marks]

End of the questions.

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